ROMAN TO INTEGER CONVERSION

Aim

To write a Java program to convert a Roman numeral into its corresponding integer value.

Description

The program reads a Roman numeral from the user and scans its characters from right to left. Each Roman symbol is converted to its numeric value. If a symbol is smaller than the previous symbol, its value is subtracted; otherwise, its value is added. The final integer value is displayed. Invalid Roman numeral characters produce an error message.

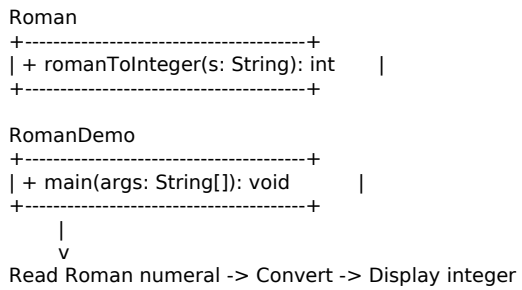
Software Requirements

Hardware: Computer or laptop. Software: Java Development Kit (JDK) and a Java IDE or text editor such as Notepad, VS Code, or Eclipse.

Algorithm

1. Start the program.
2. Read the Roman numeral from the user.
3. Convert the input to uppercase and remove extra spaces at the ends.
4. Initialize result and previous to 0.
5. Traverse the string from right to left.
6. Convert each Roman character to its numeric value using switch.
7. If the current value is smaller than previous, subtract it from result; otherwise, add it.
8. Store the current value in previous.
9. Display the integer value, or display an error for an invalid character.
10. Close the Scanner and stop.

UML Diagram



Code

```
package recordprograms;

public class Roman {

    public int romanToInteger(String s) {
        int result = 0;
        int previous = 0;

        for (int i = s.length() - 1; i >= 0; i--) {
            int value;

            switch (s.charAt(i)) {
                case 'I':
                    value = 1;
                    break;
                case 'V':
                    value = 5;
                    break;
                case 'X':
                    value = 10;
                    break;
                case 'L':
                    value = 50;
                    break;
                case 'C':
                    value = 100;
                    break;
                case 'D':
                    value = 500;
                    break;
                case 'M':
                    value = 1000;
                    break;
                default:
                    throw new IllegalArgumentException(
                        "Invalid Roman numeral character: "
                        + s.charAt(i)
                    );
            }

            if (value < previous) {
                result -= value;
            } else {
                result += value;
            }

            previous = value;
        }

        return result;
    }
}

import java.util.Scanner;
import recordprograms.Roman;

public class RomanDemo {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a Roman numeral: ");
        String input = sc.nextLine().trim().toUpperCase();

        try {
            Roman roman = new Roman();
            int result = roman.romanToInteger(input);

            System.out.println("Integer value: " + result);
        } catch (IllegalArgumentException e) {
            System.out.println(e.getMessage());
        }

        sc.close();
    }
}
```

Output Screenshot

Example 1: Valid Roman numeral

```
Enter a Roman numeral: MCMXCIV
Integer value: 1994
```

Example 2: Invalid character

```
Enter a Roman numeral: ABC
Invalid Roman numeral character: C
```

Result

Thus, the Java program to convert a Roman numeral into an integer value was prepared. The valid input example MCMXCIV produces 1994.